$C_{lq}^{(1)}\text{i1\_i2\_i3\_i4\_} \to$

$$\frac{1}{16\pi^2}\left(\frac{1}{432}\frac{1}{m_\Phi^2}\left(13 g_1^2 c_\gamma^2 \overline{y_u}^{i4p} y_u^{i3p}\delta_{i1i2} + s_\gamma^2 \overline{y_d}^{i4p} y_d^{i3p}\left(54 c_\gamma^2 \overline{y_e}^{i2r} y_e^{i1r} - 5 g_1^2 \delta_{i1i2}\right) + \right.\right.$$

$$\left.s_\gamma^2 \overline{y_e}^{i2p} y_e^{i1p}\left(54 c_\gamma^2 \overline{y_u}^{i4r} y_u^{i3r} + 7 g_1^2 \delta_{i3i4}\right)\right) - \frac{1}{81}\sum_p g_1^4 \text{LF}_{3,0}\left[m_{\tilde d}^{p}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{5}{216}\sum_p g_1^4 \text{LF}_{4,-1}\left[m_{\tilde d}^{p}\right]\delta_{i1i2}\delta_{i3i4} - \frac{4}{405}\sum_p g_1^4 \text{LF}_{5,-2}\left[m_{\tilde d}^{p}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{27}\sum_p g_1^4 \text{LF}_{3,0}\left[m_{\tilde e}^{p}\right]\delta_{i1i2}\delta_{i3i4} + \frac{5}{72}\sum_p g_1^4 \text{LF}_{4,-1}\left[m_{\tilde e}^{p}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{4}{135}\sum_p g_1^4 \text{LF}_{5,-2}\left[m_{\tilde e}^{p}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54}\sum_p g_1^4 \text{LF}_{3,0}\left[m_{\tilde l}^{p}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{5}{144}\sum_p g_1^4 \text{LF}_{4,-1}\left[m_{\tilde l}^{p}\right]\delta_{i1i2}\delta_{i3i4} - \frac{2}{135}\sum_p g_1^4 \text{LF}_{5,-2}\left[m_{\tilde l}^{p}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{162}\sum_p g_1^4 \text{LF}_{3,0}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\delta_{i3i4} + \frac{5}{432}\sum_p g_1^4 \text{LF}_{4,-1}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{2}{405}\sum_p g_1^4 \text{LF}_{5,-2}\left[m_{\tilde q}^{p}\right]\delta_{i1i2}\delta_{i3i4} - \frac{4}{81}\sum_p g_1^4 \text{LF}_{3,0}\left[m_{\tilde u}^{p}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{5}{54}\sum_p g_1^4 \text{LF}_{4,-1}\left[m_{\tilde u}^{p}\right]\delta_{i1i2}\delta_{i3i4} - \frac{16}{405}\sum_p g_1^4 \text{LF}_{5,-2}\left[m_{\tilde u}^{p}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{36}\left(4 g_1^2 c_\gamma^2 \overline{y_u}^{i4p} y_u^{i3p}\delta_{i1i2} + s_\gamma^2 \overline{y_d}^{i4p} y_d^{i3p}\left(9 c_\gamma^2 \overline{y_e}^{i2r} y_e^{i1r} - 2 g_1^2\delta_{i1i2}\right) + \right.$$

$$\left.s_\gamma^2 \overline{y_e}^{i2p} y_e^{i1p}\left(9 c_\gamma^2 \overline{y_u}^{i4r} y_u^{i3r} + 2 g_1^2 \delta_{i3i4}\right)\right)\text{LF}_{1,2}[m_\Phi] +$$

$$\frac{1}{72}\left(-3 g_1^2 c_\gamma^2 \overline{y_u}^{i4p} y_u^{i3p}\delta_{i1i2} + 3 s_\gamma^2 \overline{y_d}^{i4p} y_d^{i3p}\left(3 s_\gamma^2 \overline{y_e}^{i2r} y_e^{i1r} + g_1^2\delta_{i1i2}\right) - \right.$$

$$\left.s_\gamma^2 \overline{y_e}^{i2p} y_e^{i1p}\left(9 c_\gamma^2 \overline{y_u}^{i4r} y_u^{i3r} + g_1^2 \delta_{i3i4}\right)\right)\text{LF}_{2,1}[m_\Phi] +$$

$$\frac{1}{216} g_1^2\left(9\left(-s_\gamma^2 \overline{y_d}^{i4p} y_d^{i3p} + c_\gamma^2 \overline{y_u}^{i4p} y_u^{i3p}\right)\delta_{i1i2} + \left(3 s_\gamma^2 \overline{y_e}^{i2p} y_e^{i1p} - 4 g_1^2 \delta_{i1i2}\right)\delta_{i3i4}\right)$$

$$\text{LF}_{3,0}[m_\Phi] + \frac{5}{144} g_1^4 \text{LF}_{4,-1}[m_\Phi]\delta_{i1i2}\delta_{i3i4} - \frac{2}{135} g_1^4 \text{LF}_{5,-2}[m_\Phi]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{54} g_1^4 \text{LF}_{3,0}[\tilde\mu]\delta_{i1i2}\delta_{i3i4} - \frac{1}{36} g_1^4 \text{LF}_{4,-1}[\tilde\mu]\delta_{i1i2}\delta_{i3i4} + \frac{4}{135} g_1^4 \text{LF}_{5,-2}[\tilde\mu]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{288} g_1^4 \text{LF}_{2,1,0}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{288} g_1^4 \text{LF}_{2,2,-1}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{144} g_1^4 \text{LF}_{3,1,-1}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{288} g_1^4 \text{LF}_{4,1,-2}\left[m_1, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{288} g_1^4 \text{LF}_{2,1,0}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{288} g_1^4 \text{LF}_{2,2,-1}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{144} g_1^4 \text{LF}_{3,1,-1}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{288} g_1^4 \text{LF}_{4,1,-2}\left[m_1, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{2592} g_1^4 \text{LF}_{2,1,0}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{2592} g_1^4 \text{LF}_{2,2,-1}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{1296} g_1^4 \text{LF}_{3,1,-1}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{2592} g_1^4 \text{LF}_{4,1,-2}\left[m_1, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{2592} g_1^4 \text{LF}_{2,1,0}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{2592} g_1^4 \text{LF}_{2,2,-1}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{1296} g_1^4 \text{LF}_{3,1,-1}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{2592} g_1^4 \text{LF}_{4,1,-2}\left[m_1, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}\left[m_2, m_{\tilde l}^{i1}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}\left[m_2, m_{\tilde l}^{i2}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}\left[m_2, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{2,2,-1}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{4,1,-2}\left[m_2, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,1,0}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,2,-1}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 \text{LF}_{4,1,-2}\left[m_3, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,1,0}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 \text{LF}_{2,2,-1}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 \text{LF}_{4,1,-2}\left[m_3, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} -$$

$$\frac{1}{18} g_1^2 \overline{y_d}^{i4p} y_d^{i3p} \text{LF}_{2,1,0}\left[m_{\tilde d}^{p}, \tilde\mu\right]\delta_{i1i2} + \frac{1}{36} g_1^2 \overline{y_d}^{i4p} y_d^{i3p} \text{LF}_{2,2,-1}\left[m_{\tilde d}^{p}, \tilde\mu\right]\delta_{i1i2} +$$

$$\frac{1}{36} g_1^2 \overline{y_d}^{i4p} y_d^{i3p} \text{LF}_{3,1,-1}\left[m_{\tilde d}^{p}, \tilde\mu\right]\delta_{i1i2} + \frac{1}{18} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{2,1,0}\left[m_{\tilde e}^{p}, \tilde\mu\right]\delta_{i3i4} -$$

$$\frac{1}{36} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{2,2,-1}\left[m_{\tilde e}^{p}, \tilde\mu\right]\delta_{i3i4} - \frac{1}{36} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{3,1,-1}\left[m_{\tilde e}^{p}, \tilde\mu\right]\delta_{i3i4} +$$

$$\frac{1}{144} g_1^4 \text{LF}_{2,1,0}\left[m_{\tilde l}^{i1}, m_1\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{288} g_1^4 \text{LF}_{3,1,-1}\left[m_{\tilde l}^{i1}, m_1\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_{\tilde l}^{i1}, m_2\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_{\tilde l}^{i1}, m_2\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{144} g_1^4 \text{LF}_{2,1,0}\left[m_{\tilde l}^{i2}, m_1\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{288} g_1^4 \text{LF}_{3,1,-1}\left[m_{\tilde l}^{i2}, m_1\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_{\tilde l}^{i2}, m_2\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_{\tilde l}^{i2}, m_2\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{1296} g_1^4 \text{LF}_{2,1,0}\left[m_{\tilde q}^{i3}, m_1\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{2592} g_1^4 \text{LF}_{3,1,-1}\left[m_{\tilde q}^{i3}, m_1\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_{\tilde q}^{i3}, m_2\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_{\tilde q}^{i3}, m_2\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}\left[m_{\tilde q}^{i3}, m_3\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 \text{LF}_{3,1,-1}\left[m_{\tilde q}^{i3}, m_3\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{1296} g_1^4 \text{LF}_{2,1,0}\left[m_{\tilde q}^{i4}, m_1\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{2592} g_1^4 \text{LF}_{3,1,-1}\left[m_{\tilde q}^{i4}, m_1\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{48} g_1^2 g_2^2 \text{LF}_{2,1,0}\left[m_{\tilde q}^{i4}, m_2\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{96} g_1^2 g_2^2 \text{LF}_{3,1,-1}\left[m_{\tilde q}^{i4}, m_2\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}\left[m_{\tilde q}^{i4}, m_3\right]\delta_{i1i2}\delta_{i3i4} - \frac{1}{54} g_1^2 g_3^2 \text{LF}_{3,1,-1}\left[m_{\tilde q}^{i4}, m_3\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{9} g_1^2 \overline{y_u}^{i4p} y_u^{i3p} \text{LF}_{2,1,0}\left[m_{\tilde u}^{p}, \tilde\mu\right]\delta_{i1i2} - \frac{1}{18} g_1^2 \overline{y_u}^{i4p} y_u^{i3p} \text{LF}_{2,2,-1}\left[m_{\tilde u}^{p}, \tilde\mu\right]\delta_{i1i2} -$$

$$\frac{1}{18} g_1^2 \overline{y_u}^{i4p} y_u^{i3p} \text{LF}_{3,1,-1}\left[m_{\tilde u}^{p}, \tilde\mu\right]\delta_{i1i2} + \frac{1}{36} g_1^2 \overline{y_d}^{i4p} y_d^{i3p} \text{LF}_{2,1,0}\left[\tilde\mu, m_{\tilde d}^{p}\right]\delta_{i1i2} +$$

$$\frac{5}{72} g_1^2 \overline{y_d}^{i4p} y_d^{i3p} \text{LF}_{3,1,-1}\left[\tilde\mu, m_{\tilde d}^{p}\right]\delta_{i1i2} - \frac{1}{72} g_1^2 \overline{y_d}^{i4p} y_d^{i3p} \text{LF}_{4,1,-2}\left[\tilde\mu, m_{\tilde d}^{p}\right]\delta_{i1i2} -$$

$$\frac{1}{36} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{2,1,0}\left[\tilde\mu, m_{\tilde e}^{p}\right]\delta_{i3i4} + \frac{1}{72} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{3,1,-1}\left[\tilde\mu, m_{\tilde e}^{p}\right]\delta_{i3i4} -$$

$$\frac{1}{72} g_1^2 \overline{y_e}^{i2p} y_e^{i1p} \text{LF}_{4,1,-2}\left[\tilde\mu, m_{\tilde e}^{p}\right]\delta_{i3i4} - \frac{1}{18} g_1^2 \overline{y_u}^{i4p} y_u^{i3p} \text{LF}_{2,1,0}\left[\tilde\mu, m_{\tilde u}^{p}\right]\delta_{i1i2} -$$

$$\frac{1}{72} g_1^2 \overline{y_u}^{i4p} y_u^{i3p} \text{LF}_{3,1,-1}\left[\tilde\mu, m_{\tilde u}^{p}\right]\delta_{i1i2} - \frac{1}{72} g_1^2 \overline{y_u}^{i4p} y_u^{i3p} \text{LF}_{4,1,-2}\left[\tilde\mu, m_{\tilde u}^{p}\right]\delta_{i1i2} +$$

$$\frac{1}{288} g_1^4 \text{LF}_{2,1,1,-1}\left[m_1, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} + \frac{1}{144} g_1^4 m_1^2 \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{288} g_1^4 \text{LF}_{2,1,1,-1}\left[m_1, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} + \frac{1}{144} g_1^4 m_1^2 \text{LF}_{2,1,1,0}\left[m_1, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{3}{32} g_2^4 \text{LF}_{2,1,1,-1}\left[m_2, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} + \frac{3}{16} g_2^4 m_2^2 \text{LF}_{2,1,1,0}\left[m_2, m_{\tilde l}^{i1}, m_{\tilde q}^{i3}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{3}{32} g_2^4 \text{LF}_{2,1,1,-1}\left[m_2, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} + \frac{3}{16} g_2^4 m_2^2 \text{LF}_{2,1,1,0}\left[m_2, m_{\tilde l}^{i2}, m_{\tilde q}^{i4}\right]\delta_{i1i2}\delta_{i3i4} +$$

$$\frac{1}{8}\overline{y_d}^{i4p} y_d^{i3p}\overline{y_e}^{i2r} y_e^{i1r} \text{LF}_{2,1,1,-1}\left[\tilde\mu, m_{\tilde d}^{p}, m_{\tilde e}^{r}\right] +$$

$$\left.\frac{1}{4}\tilde\mu^2\overline{y_e}^{i2p} y_e^{i1p}\overline{y_u}^{i4r} y_u^{i3r} \text{LF}_{2,1,1,0}\left[\tilde\mu, m_{\tilde e}^{p}, m_{\tilde u}^{r}\right]\right)$$